Sec: SR.IIT_*COSC(MODEL-A&APEX) | GTM-6 | Date: 15-03-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**B**	2	**A**	3	**D**	4	**C**	5	**B**
6	**A**	7	**B**	8	**C**	9	**D**	10	**D**
11	**B**	12	**A**	13	**A**	14	**D**	15	**B**
16	**B**	17	**A**	18	**C**	19	**B**	20	**B**
21	**2**	22	**22**	23	**180**	24	**1**	25	**2**

CHEMISTRY

26	**A**	27	**D**	28	**C**	29	**C**	30	**C**
31	**D**	32	**C**	33	**C**	34	**A**	35	**C**
36	**C**	37	**B**	38	**D**	39	**A**	40	**D**
41	**B**	42	**C**	43	**B**	44	**A**	45	**B**
46	**2**	47	**57**	48	**52**	49	**4**	50	**6**

MATHEMATICS

51	**C**	52	**C**	53	**A**	54	**B**	55	**B**
56	**C**	57	**B**	58	**D**	59	**B**	60	**C**
61	**B**	62	**C**	63	**B**	64	**C**	65	**C**
66	**A**	67	**A**	68	**D**	69	**C**	70	**D**
71	**1**	72	**8**	73	**4**	74	**2**	75	**6**

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SOLUTIONS

PHYSICS

1. P is a capacitor and Q is resister. Impedance of PQ, $Z+\sqrt{X_C^2+R^2}$

   Current lead by $\tan^{-1}\left(\frac{X_C}{R}\right)$

2. Buoyancy is a real force. It is independent of frame.
3. Every distance increases as we heat the plate.
4. 2 beats with 514Hz= 512 and 516 are possible
   6 beats with 510Hz= 504 and 516 are possible
   So, actual frequency of fork shall be 516Hz
5. $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$ using this we get image distance as 30cm.
6. $\frac{P}{Q}=\frac{40}{60},\frac{P}{\frac{10Q}{10+Q}}=\frac{50}{50}=1$

   Solving $P=\frac{10}{3}\Omega, Q=5\Omega$
7. Draw truth table, we get $Y=A\overline{B}$
8. $-W_{Gravity}=W_{person}+W_{spring}=W+E=\Delta PE$
9. BD is dimensionless. C must have same dimensions as AD.
10. $\omega\,\alpha\,\frac{v}{r}\,\alpha\,\frac{z/n}{n^2/z}$

    $T=\frac{2\pi}{\omega}$

    $mvr=\frac{nh}{2\pi}$
11. The effective length if pendulum increases as the water leaks because COM lowers.
12. $\lambda_m T=cons\tan t$

    $power\ \alpha T^4$
13. $r<r_A, E=0, r>r_A$ Field shall be away from centre and for $r>r_B$, field shall be towards centre.
14. Energy radiated by sun = $4\pi r^2\left(\text{int}ensity\ at\ earth\right)\times t$

    $mc^2=Energy\ radiated$
15. Total momentum delivered = $\frac{\left(Intensity\right)Area\times time}{hc/\lambda}\left(\frac{h}{\lambda}\right)$
16. $a=\left(g\sin 30\right)\left(\sin 30\right)$

    $\frac{1}{2}at^2=OA=5$

    $t=2$
17. $i_1=i_0e^{\frac{-t}{RC}}, i_2=i_s\left(1-e^{\frac{-Rt}{L}}\right)$

    $i=i_1+i_2$

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18. $for\ r < R, U = -\dfrac{GMm}{R}$

$for\ r > R, U = -\dfrac{GMm}{r}$

19. $mv_0 = mv_{cm}$

$mv_0 x = \dfrac{ml^2}{12}\omega$

$\dfrac{l\omega}{2} - v_{cm} = 0$

20. Path difference $= \int_0^l (1+ax)\,dx - l = (2n-1)\dfrac{\lambda}{2}$

21. $u\sin\theta t - \dfrac{1}{2}gt^2 = 15$

solving we get $t = 1, t = 3$

22. $E = \dfrac{Q_2}{C_2} + \dfrac{Q_2 - Q_1}{C_2}$

$E = \dfrac{Q_1 - Q_2}{C_2} + \dfrac{Q_1}{C_1}$

solving we get $\dfrac{Q_2 - Q_1}{C_2} = 22$

23. Due to perpendicular mirrors, the total deviation suffered will be 180°

24. $\tau = NiAB\sin(60^\circ) = (2.4)^2(3)^{\frac{1}{2}}$

25. Let B travel along x-direction and common normal make angle θ with x-direction
A travels along common normal after collision.

$m_B v = m_A v_A \cos\theta\ (for\ x-direction)$

$for\ m_B\dfrac{v}{2} = m_A v_A \sin\theta\ (for\ y-direction)$

$2 = \cot\theta$

$\tan\theta = \dfrac{1}{2}$

$\theta = \tan^{-1}\left(\dfrac{1}{2}\right)$

CHEMISTRY

27. $H_3PO_4 \longrightarrow 3H^+ + PO_4^{3-}$

$1-x \qquad 3x \quad x$

$\dfrac{3x+x}{1-x} = \dfrac{8}{3}; x = \dfrac{8}{20} = 0.4$

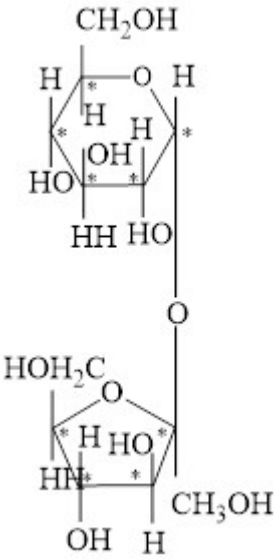
$\%\ \text{Dissociation} = \dfrac{x}{1}\times 100 = \dfrac{0.4}{1}\times 100 = 40\%$

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27. $\Delta_r S_u^o = \sum S_P - \sum S_R$

$= cS_C^o + dS_C^o - 0 - 0 = cS_C^o + dS_C^o$

28.



29. $n = 4$, total radial nodes = 3 + 2 + 1 + 0 = 6

30. On doubling the volume concentration decreases

31. In option A, B, C carbocation is stable so rate of S_N^1 is correct but in D due to resonance stabilized transition state S_N^2 rate is high

32. A) $\Delta G^o_{Fe^{3+}/Fe} = \Delta G^o_{Fe^{3+}/Fe^{2+}} + \Delta G^o_{Fe^{2+}/Fe}$

$\Rightarrow -3 \times FE^o_{(Fe^{+3}/Fe)} = -1 \times FE^o_{(Fe^{+3}/Fe^{+2})} + \left(-2 \times FE^o_{Fe^{+2}/Fe}\right)$

$\Rightarrow FE^o_{Fe^{+3}/Fe} = -0.04V$

B) $O_2(g) + 2H_2O + 4e^- \longrightarrow 4\bar{O}H \quad E^o = 0.40\,V ---(i)$

$2H_2O \longrightarrow O_2(g) + 4H^+ + 4e^- \quad E^o = -1.23\,V ----(ii)$

So $4H_2O \quad 4H^+ + 4\bar{O}H -----(iii)$

E^o For IIIrd reduction = $0.40 - 1.23 = -0.83\,V$

C) $\Delta G^o_{(Cu^{+2}/Cu)} = -1 \times FE^o_{Cu^{+2}/Cu^+} + \left(-1 \times F \times E^o_{Cu^+/Cu}\right)$

$-2 \times FE^o_{Cu^{+2}/Cu} = -1 \times FE^o_{Cu^{+2}/Cu^+} + \left(-1 \times F \times E^o_{Cu^+/Cu}\right)$

$\Rightarrow E^o_{Cu^{+2}/Cu} = -0.18V$

D) $\Delta G^o_{Cr^{+2}/Cr^{+2}} = \Delta G^o_{Cr^{+3}/Cr} + \Delta G^o_{Cr/Cr^{+2}}$

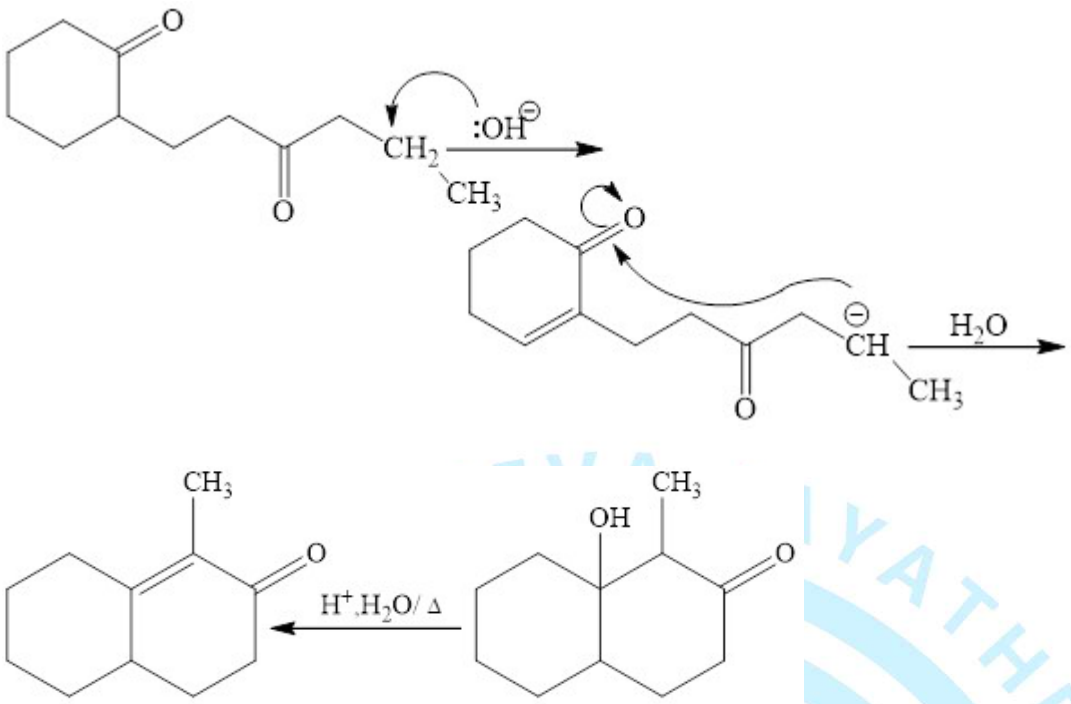
$-1 \times F \times E^o_{Cr^{+3}/Cr^{+2}} = -3 \times F \times E^o_{Cr^{+3}/Cr} + \left(-2 \times F \times E^o_{Cr/Cr^{+2}}\right)$

$\Rightarrow E^o_{Cr^{+3}/Cr^{+2}} = -0.4\,V$

33. Bromine oxidises only iodide

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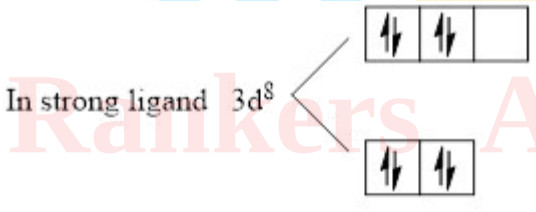
34.

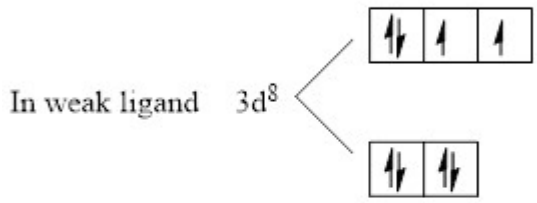


35. Conceptual

36. $\left[NiCl_2\ P(C_2H_5)_2(C_6H_5)_2\right]^{+2}$

$Ni^{+2}(28):[Ar]4s^o3d^8$





Hybridisation is dsp^2 (diamagnetic)

→ Square planar

Hybridisation is sp^3 (paramagnetic)

→ tetrahedral

37. B) III > II > I More the branching in an alkane, lesser will be the surface area, lesser will be the boiling point

38. Overall order of reaction can be decided by the data given $t_{75\%} = 2t_{50\%}$

∴ It is a first order reaction with respect to P

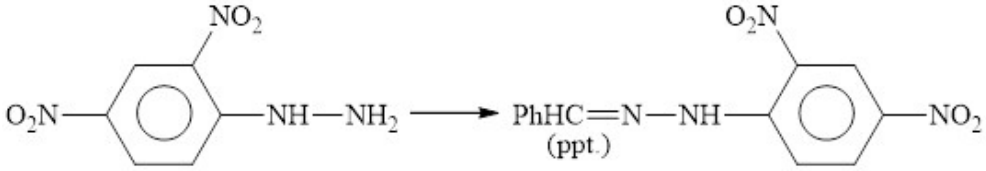
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From graph $[Q]$ is linearly decreasing with time, i.e., order of reaction with respect to Q is zero and the rate expression is $r = k[P]^1[Q]^0$

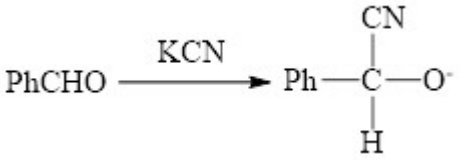
39. A) $A \to 1,4; B \to 2; C \to 2,3,4; D \to 2,3$

(Note: Assuming $AgNO_3$ is ammonical)

A)

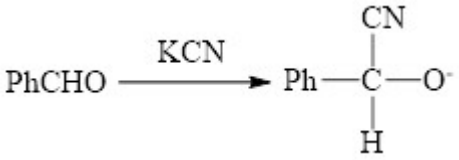


$$PhCHO + Ag_2O \xrightarrow{NH_3} PhCOO^- + \underset{(\text{white ppt})}{Ag\downarrow}$$



B) $CH_3C \equiv CH \xrightarrow{\text{ammonical } AgNO_3} \underset{(\text{White ppt})}{CH_3 - C \equiv C^- Ag^+ \downarrow}$

C)



$$AgNO_3 + CN^- \longrightarrow AgCN\downarrow$$

D) $AgNO_3 + I^- \longrightarrow AgI\downarrow$

40. (D): $C=C$ at position 1 is more reactive towards electrophile $Br^{\oplus}$ as this is adjacent to O atom to provide +M effect. $C=C$ at position 3 is more substituted as compared to $C=C$ at position 2. $C \equiv C$ is less reactive towards $Br^{\oplus}$ as compared to $C=C$

41. (B) : I < II – Due to ortho effect of - CH_3 group (steric inhibition to resonance)

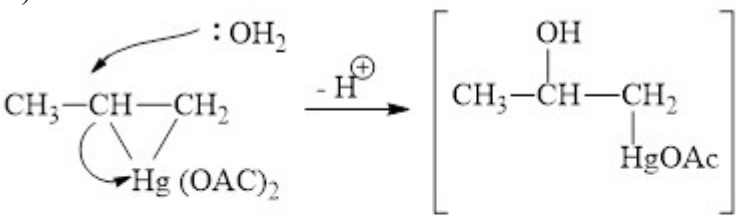
III > IV – Due to +I effect of CH_3 group

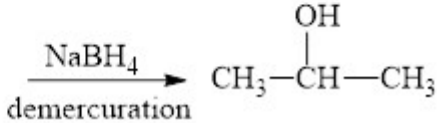
VII > VIII – Steric hindrance to solvation by - CH_3 group in conjugate acid in o-toluidine makes it a weaker base than aniline

42.

A) $(CH_3CH_2CH_2-)_3B \xrightarrow{HO^{\ominus}} 3CH_3CH_2CH_2OH + H_3BO_3$

B)





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C)

$$CH_3-CH=CHMgBr \xrightarrow[(ii)\ H_2O]{(i)\ O_2,\ \Delta} CH_3-CH=CH-OH \xrightarrow{\text{tautomerises}} CH_3-CH_2-CHO$$

D)

$$CH_3-CH_2-CH_2MgBr + \underset{\diagdown O \diagup}{CH_2-CH_2} \xrightarrow{(THF)} (CH_3CH_2CH_2CH_2CH_2OMgBr) \xrightarrow{H_3O^+} CH_3CH_2CH_2CH_2CH_2OH + Mg(OH)Br$$

43. (B): (A) Formula silver chromate will be Ag_2CrO_4

B) Minimum mass of zinc required for reduction of 0.1 mole of Cr^{3+} to Cr^{2+}

$= \frac{0.1}{2}$ moles of $Zn = \frac{6.54}{2}g = 3.27g$

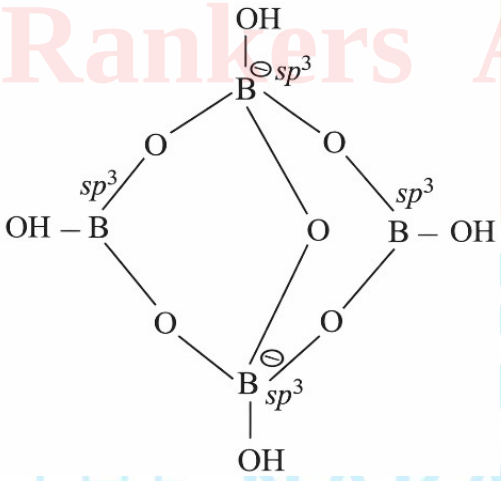
C) $CrO_4^{2-} \rightleftharpoons Cr_2O_7^{2-}$

in both ions chromium is in + 6 state

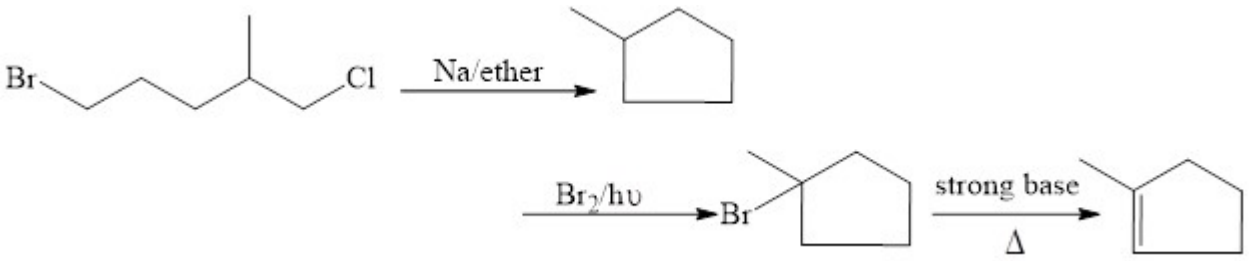
D) Given reaction is correct

44. The correct formula of borax is $Na_2\left[B_4O_5(OH)_4\right]8H_2O$

The structure of anion is



45.



46. Conceptual.

47. $P_4 + 5O_2 \rightarrow P_4O_{10}$

0.2 0.2

Mass of $P_4O_{10} = 0.2 \times 284 = 56.8$

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48. $PO_4^{3-} \xrightarrow[+HNO_3]{(NH_4)_2MoO_4,\ T\le 40^\circ C} \underset{\text{Caneiry yellow ppt.}}{(NH_4)_3PO_4.12MoO_4}$

52 o-atoms are present in product

49. $BeCl_2 \rightarrow sp \rightarrow$ linear

$N_3^- \rightarrow sp \rightarrow$ linear

$N_2O \rightarrow sp \rightarrow$ linear

$\overset{\oplus}{N}O_2 \rightarrow sp \rightarrow$ linear

$O_3 \rightarrow sp^2 \rightarrow$ linear

$SCl_2 \rightarrow sp^3 \rightarrow$ linear

$I_3^- \rightarrow sp^3d \rightarrow$ linear

$ICl_2^- \rightarrow sp^3d \rightarrow$ linear

$XeF_2 \rightarrow sp^3d \rightarrow$ linear

Among the following only four (4) has linear shape and no d-orbital is involved in hybridization

50. Let number of glycine units = n

Mass of decapeptide = 796

Mass of H_2O needed = 162 g, Total mass = 958 g

$$958 \times \frac{47}{100} = 75 \times n$$

$$\therefore n = \frac{958 \times 47}{100 \times 75} \approx 6$$

MATHS

51. Let e_1 be eccentricity of ellipse

$$\Rightarrow e_1 = \sqrt{1 - \frac{16}{36}} = \sqrt{1 - \frac{4}{9}} = \frac{\sqrt{5}}{3}$$

So, $ae_1 = 6 \cdot \frac{\sqrt{5}}{3} = 2\sqrt{5}$

Now, H: $\frac{x^2}{p^2} - \frac{y^2}{q^2} = 1.$

$$p \cdot e = ae_1 \Rightarrow p \cdot 5 = 2\sqrt{5}$$

$$p = \frac{2}{\sqrt{5}} \Rightarrow e^2 = 1 + \frac{q^2}{p^2}$$

$$\Rightarrow 25 = 1 + \frac{5q^2}{4} \Rightarrow q^2 = \frac{96}{5}.$$

61. $2[x] = \{x+5\} + x - 1 \Rightarrow 2[x] = \{x\} + x - 1 \Rightarrow x = -1, \frac{1}{2}$

62. $2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = 0$

$\Rightarrow (2\vec{a} + 3\vec{d}) \times \vec{c} = 0 \Rightarrow \vec{c} = \lambda(2\vec{a} + 3\vec{d})$

$\Rightarrow \vec{c} = \lambda(7i - 13j + 19k)$

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Now $\left(\vec{a}-\vec{b}\right).\vec{c}--=\lambda\left(7+52+38\right)+97\lambda=-97$

$\Rightarrow\lambda=-1$

Now $\vec{c}=-7i+13j-19k\Rightarrow\vec{c}\times\vec{k}--=7j+13i\Rightarrow\left|\vec{c}\times\hat{k}\right|^2=7^2+13^2=218$

63. $f^{-1}\left(x\right)=x-1$

$\Rightarrow h^{-1}\left(x\right)=x^3+x$

$\therefore\frac{d}{dx}\left(h^{-1}\left(x\right)\right)=3x^2+1$

64. $f\left(x\right)=\frac{e^x}{x^2}$

65. Given $P=\left(1,2\right)$

Using the parametric from, the coordinates of the point at a distance of $\frac{\sqrt{6}}{3}$ i.e. $\sqrt{\frac{2}{3}}$ is given by

$\left(1+\sqrt{\frac{2}{3}}\cos\alpha,2+\sqrt{\frac{2}{3}}\sin\theta\right)$

This point lies on the line $x+y=4$

$\Rightarrow\left(1+\sqrt{\frac{2}{3}}\cos\theta\right)+\left(2+\sqrt{\frac{2}{3}}\sin\theta\right)=4$

$\Rightarrow3+\sqrt{\frac{2}{3}}\left(\cos\theta+\sin\theta\right)=4$

$\Rightarrow\sqrt{\frac{2}{3}}\left(\cos\theta+\sin\theta\right)=1$

Now, squaring on both sides, we get

$\frac{2}{3}\left(1+\sin2\theta\right)=1$

$\Rightarrow1+\sin\theta=\frac{3}{2}\Rightarrow\sin2\theta=\frac{1}{2}$

$\Rightarrow\theta=\frac{1}{2}\sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{12}\text{and}\frac{5\pi}{12}$

Then, the angles made by L_1,L_2 with positive X-axis are $\frac{\pi}{12}\text{and}\frac{5\pi}{12}$

66. $m_{AB}\times m_{BC}=-1$

$\Rightarrow\left(t_1+t_2\right)\left(t_2+t_3\right)=-4---\left(1\right)$

Similarly $m_{AD}\times m_{CD}=-1$

$\Rightarrow\left(t_1+t_4\right)\left(t_3+t_4\right)=-4----\left(2\right)$

Using 1 and 2 $\Rightarrow\left(t_1+t_2\right)\left(t_2+t_3\right)=\left(t_1+t_4\right)\left(t_3+t_4\right)$

Simplify we get $\frac{t_2+t_4}{t_1+t_3}=-1$

67. For infinitely many solution

$\Delta=0$

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$$\begin{vmatrix}1 & 5 & -1\\ 4 & 3 & -3\\ 24 & 1 & \lambda\end{vmatrix}=0$$

$\Rightarrow 1(3\lambda+3)-5(4\lambda+72)-1(4-72)=0$

$\Rightarrow -17\lambda+3-4\times 72=0$

$\Rightarrow 17\lambda=-289$

$\Rightarrow \lambda=-17$

$\Delta 1=0$

$$\Rightarrow\begin{vmatrix}1 & 5 & -1\\ 7 & 3 & -3\\ \mu & 1 & -17\end{vmatrix}=0$$

$\Rightarrow 1(51+3)-5(-119+3\mu)-1(7-3\mu)=0$

$\Rightarrow -48+595-15\mu-7+3\mu=0$

$\Rightarrow 12\mu=540$

$\mu=45$

$x+5y-z=1$

$4x+3y-3x=7$

$24x+y-17z=45$

Let $z=1$

$x+5y=[1+\lambda]\times 4$

$4x+3y=7+3\lambda$

$4x+20y=-4+4\lambda$

$\frac{-17y=3-\lambda}{-17}$

$y=\frac{\lambda-3}{17}, x=1+\lambda-\frac{5\lambda-15}{17}$

$=\frac{32-12\lambda}{17}$

$7\le\frac{\lambda-3}{17}+\frac{32+12\lambda}{17}+\lambda\le 77$

$7\le\frac{30\lambda+29}{17}\le 77$

$3\le\lambda\le 42$

$\lambda=3,20,37$

68. $f'(x)=ax(x-1)\Rightarrow f'(2)=6\Rightarrow a=3$

$f'(x)=3(x^2-x)\Rightarrow f(x)=x^3-\frac{3x^2}{2}+C$

$\therefore f(x)=x^2\left(x-\frac{3}{2}\right)$

$f(2)=2\Rightarrow C=0$

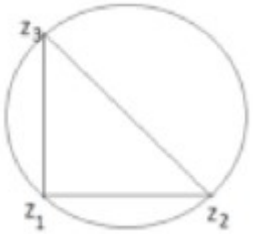
69. $P(\text{T or L})=0.5=O(E)\text{say}$

$P(\text{WWE or WEW or EWW or WWW})=4.\frac{5}{10}.\frac{5}{10}.\frac{5}{10}=\frac{1}{2}$

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70. $|z_1-0|=|z_2-0|=|z_3-0|=|z_1+z_2+z_3-0|=3$

O is the circumcentre and $z_1+z_2+z_3$ is the orthocentre $z_2+z_3=0$



$|z_1+z_2|.|z_1+z_3|=|z_1-z_3|=\left(\frac{6}{\sqrt{2}}\right)^2=18$

71.

72.

73. Assuming common difference of $[a_n]$ is d where $d\in I$

$\Rightarrow a_1+a_2=a_k$ for some positive integer k

$\Rightarrow 2a_1+d=a_1+(k-1)d\Rightarrow a_1=(k-2)d$

$\Rightarrow k\neq 2$ and $d=\frac{a_1}{(k-2)}$ as $a_n=a_1+(n-1)d\Rightarrow a_n=a_1+\left(\frac{n-1}{n-2}\right)a_1$

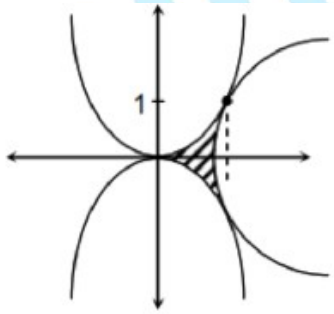
$\Rightarrow a_1+a_2+a_3+...+a_{n-2}+a_{n-1}+a_n=a_1n+\frac{n(n-1)}{2}d$

$\Rightarrow \sum_{i=1}^{n}a_i=a_1+\left((n-1)(k-2)+\frac{n(n-1)}{2}\right)d$

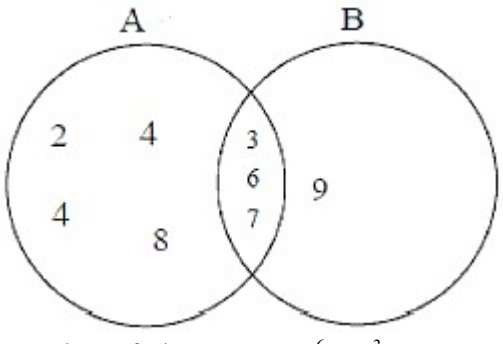
As 2019 is product of points (3) and (673)

Hence $k-2=-1,1,3,673,2019$

74. Area $=2\int\left(\frac{y^2+3}{4}-\sqrt{y}\right)dy=\frac{1}{3}$



75.



Number of elements $=2^6-2^3=64-8=56$

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